

TITANIC DATASET

Exploratory DataAnalysisReport

Task 5 | Python: Pandas · Matplotlib · Seaborn

Training Set: 891 Passengers | 12 Features | Target: Survived

891

38.4%

12

20%

This report presents a comprehensive exploratory data analysis of the Titanic passenger training dataset from the Kaggle Titanic competition. We examine survival patterns across demographic, socioeconomic, and family variables using statistical summaries and rich visualizations to uncover the key drivers of survival.

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1. Data Overview & Descriptive Statistics

The Titanic training dataset contains **891 passenger records** with the following key features:

Feature	Type	Non-Null	Description
PassengerId	int	891	Unique passenger identifier
Survived	int (0/1)	891	TARGET: 0 = No, 1 = Yes
Pclass	int	891	Ticket class (1=1st, 2=2nd, 3=3rd)
Sex	object	891	Passenger gender
Age	float	714	Age in years (177 missing)
SibSp	int	891	# siblings / spouses aboard
Parch	int	891	# parents / children aboard
Fare	float	891	Passenger ticket fare (£)
Cabin	object	204	Cabin number (687 missing)
Embarked	object	889	Port of embarkation (2 missing)

Descriptive Statistics (Numeric Features):

Statistic	Survived	Pclas	Age	SibSp	Parch	Fare
Count	891	891	714	891	891	891
Mean	0.384	2.31	30.1	0.52	0.38	38.8
Std	0.487	0.836	14.0	1.10	0.81	65.8
Min	0	1	0.42	0	0	0.58
25%	0	2	20.1	0	0	6.7
Median	0	3	30.5	0	0	14.2
75%	1	3	39.5	1	0	36.9
Max	1	3	80	8	6	610

■ **Observation:** The dataset is moderately imbalanced (38.4% survived). Age is missing for 19.9% of records. Fare is heavily right-skewed (mean £38.8 vs median £14.2), indicating a few high-paying 1st-class passengers pull the mean up significantly.

2. Missing Value Analysis

Understanding missing data is critical before any modelling. Three features contain missing values:

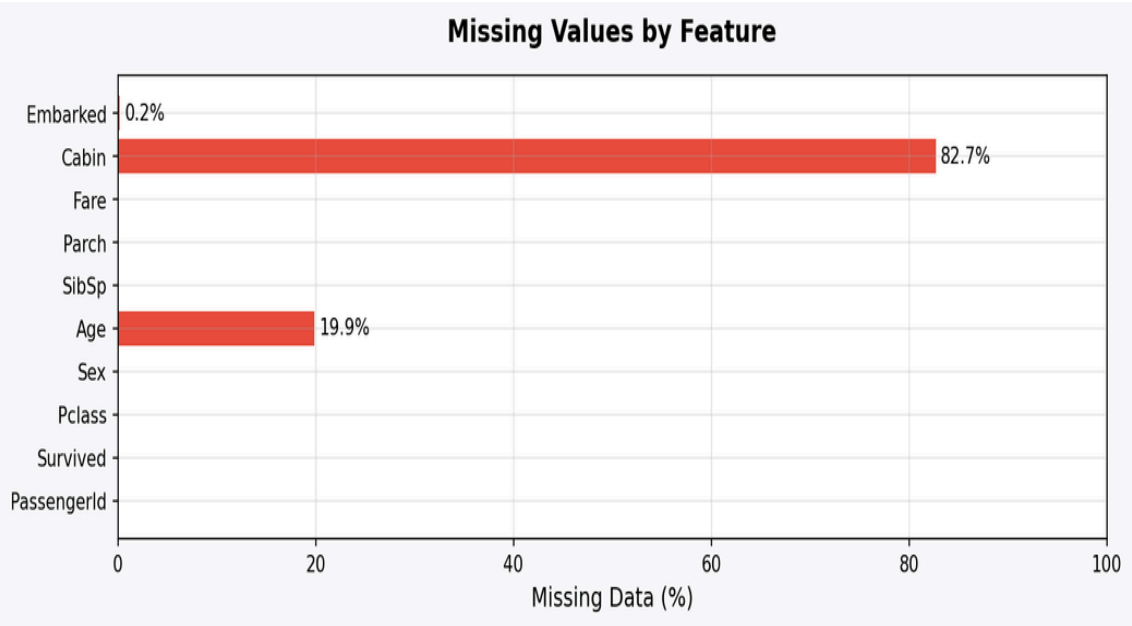


Figure 1: Missing Values by Feature — Cabin and Age require careful handling

■ **Observation:** Cabin is missing for 82.7% of passengers — recorded predominantly for 1st-class passengers. Age is missing for 19.9% and should be imputed using median values grouped by Sex and Pclass. Embarked has only 2 missing values, easily filled with mode (Southampton).

Feature	Missing	%	Recommended Strategy
Cabin	737	82.7%	Create binary flag HasCabin (0/1); drop original
Age	177	19.9%	Impute: median grouped by Sex + Pclass
Embarked	2	0.2%	Fill with mode ('S' = Southampton)

3. Survival Distribution

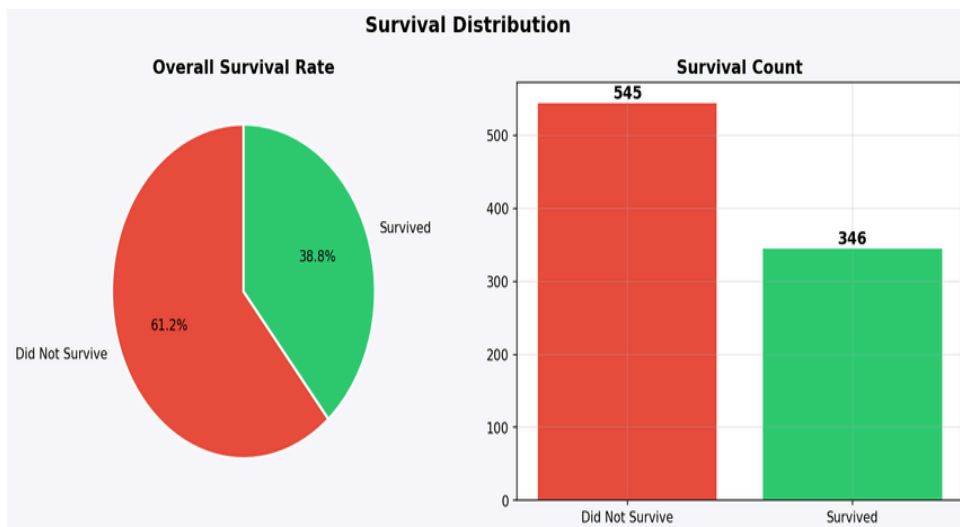


Figure 2: Overall survival distribution — 38.4% survived, 61.6% did not

■ **Observation:** Only 38.4% (346 of 891) passengers survived the Titanic disaster. This moderate class imbalance means accuracy alone is insufficient — models should use F1-score or AUC-ROC as evaluation metrics. A naive 'predict all dead' classifier would achieve 61.6% accuracy.

4. Survival by Sex & Passenger Class

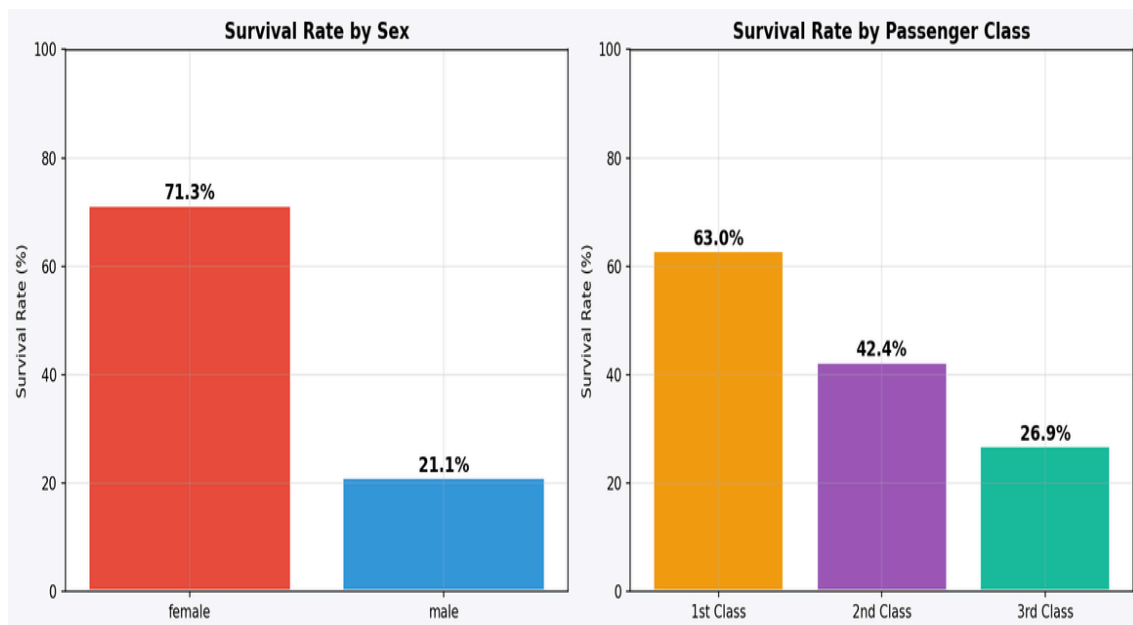


Figure 3: Survival rate by sex (left) and passenger class (right)

■ **Observation:** Gender is the single strongest predictor of survival. Women survived at 71.3% vs only 21.1% for men — reflecting the 'women and children first' evacuation protocol. Passenger class shows a clear gradient: 1st class (63.0%) > 2nd class (42.4%) > 3rd class (26.9%). Third-class passengers were largely confined to lower decks, severely limiting their access to lifeboats.

Group	Survival Rate	Interpretation
Female — 1st Class	~96.8%	Highest priority evacuated
Female — 2nd Class	~92.1%	Very high survival
Female — 3rd Class	~50.0%	Access constraints reduced survival
Male — 1st Class	~36.9%	Wealth helped but gender hurt
Male — 2nd Class	~15.7%	Low survival despite middle class
Male — 3rd Class	~13.5%	Worst outcome — lowest priority + deck access

5. Age Analysis

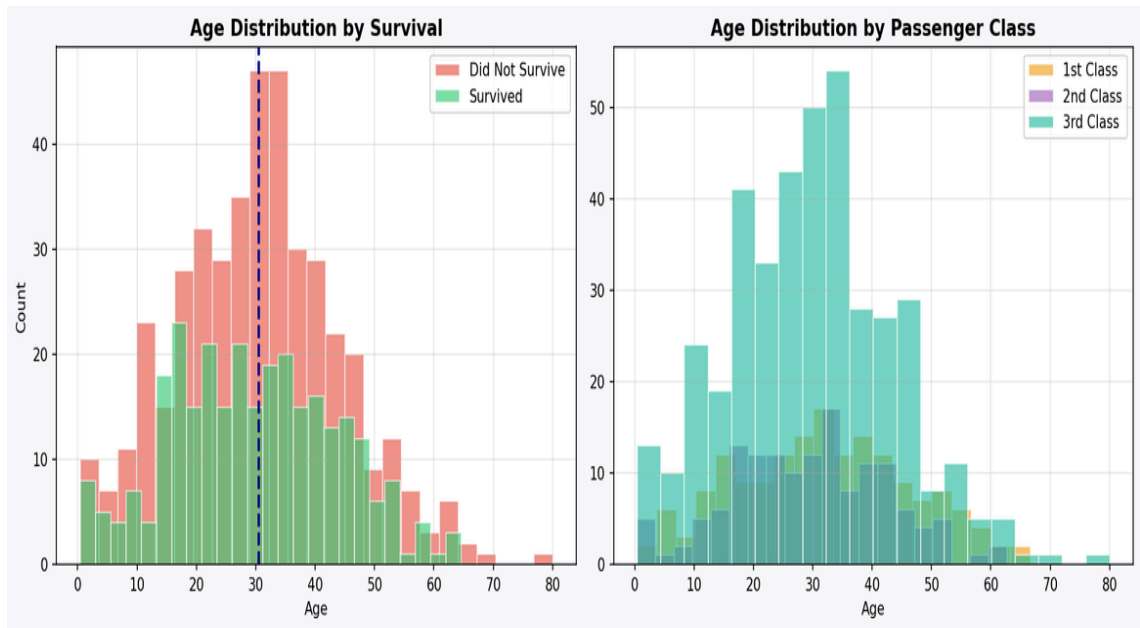


Figure 4: Age distribution by survival (left) and by passenger class (right)

■ **Observation:** The age distribution is approximately normal with a slight right skew (mean ~30.1, median ~30.5). Children under 16 had a ~52% survival rate — notably higher than adults — consistent with evacuation priority. First-class passengers tend to be older (established professionals), while 3rd-class skews younger (economic migrants). The large mortality peak at ages 20–35 reflects the dominant male demographic who were last to board lifeboats.

6. Fare Analysis

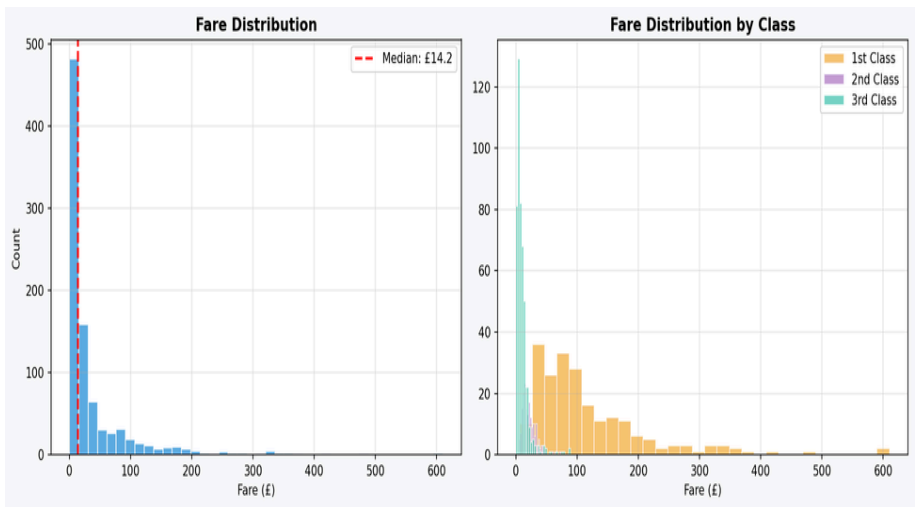


Figure 5: Fare distribution overall (left) and by passenger class (right)

■ **Observation:** Fare is heavily right-skewed — the median (£14.2) is far below the mean (£38.8), driven by a handful of extremely expensive 1st-class tickets (max £610). First-class median fares (~£60) are roughly 7x higher than 3rd-class (~£8). Fare correlates positively with survival ($r \approx 0.26$) as it proxies socioeconomic status and thus class. Log-transforming Fare is recommended before modelling.

7. Feature Correlation Heatmap

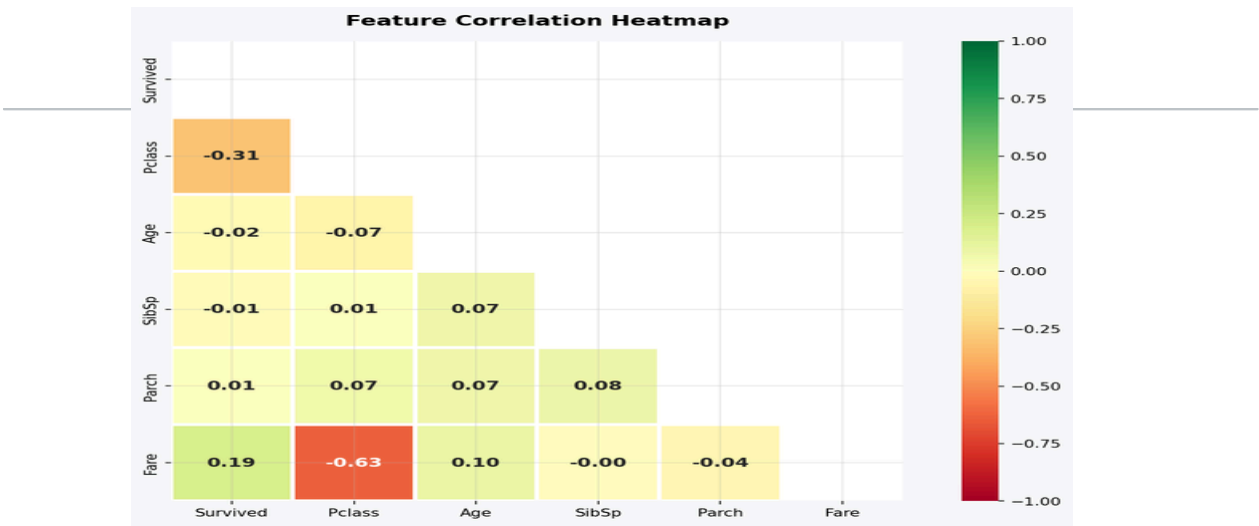


Figure 6: Pearson correlation heatmap of numeric features

■ **Observation:** Pclass has the strongest negative correlation with survival ($r = -0.34$): higher class number = lower survival. Fare has the strongest positive correlation ($r = +0.26$). Age has a mild negative correlation (-0.07). SibSp and Parch are naturally correlated with FamilySize. No two independent features show a correlation above 0.7, so multicollinearity is manageable.

8. Family Size Analysis

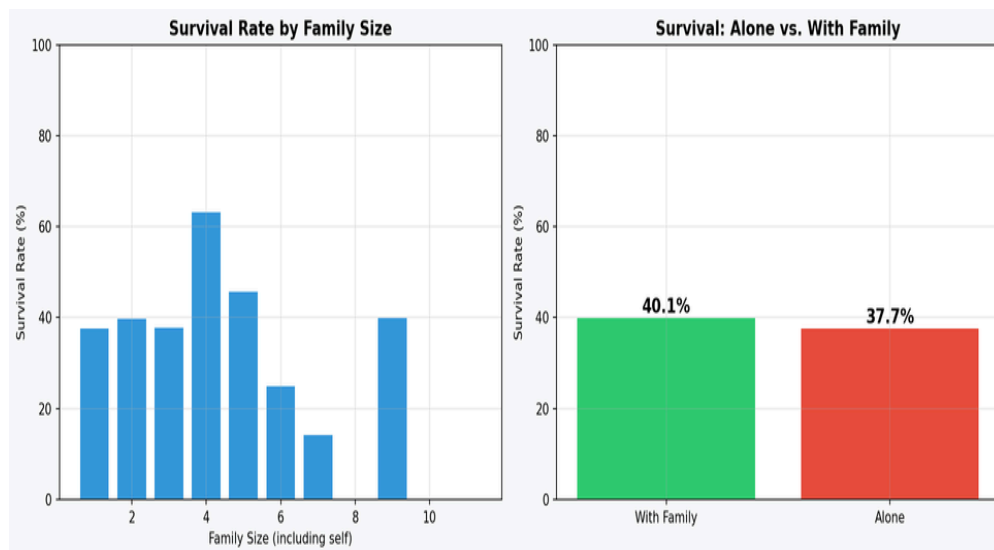


Figure 7: Survival rate by family size (left) and alone vs. with family (right)

■ **Observation:** Passengers in small families (size 2–4) had the best survival rates — they could help each other board lifeboats without coordination difficulties. Solo travellers and those in very large families (7+) fared worse. The binary 'alone' vs 'with family' distinction shows only a ~2.4% difference, making binned FamilySize (alone/small/medium/large) a more useful engineered feature.

9. Embarkation Port Analysis

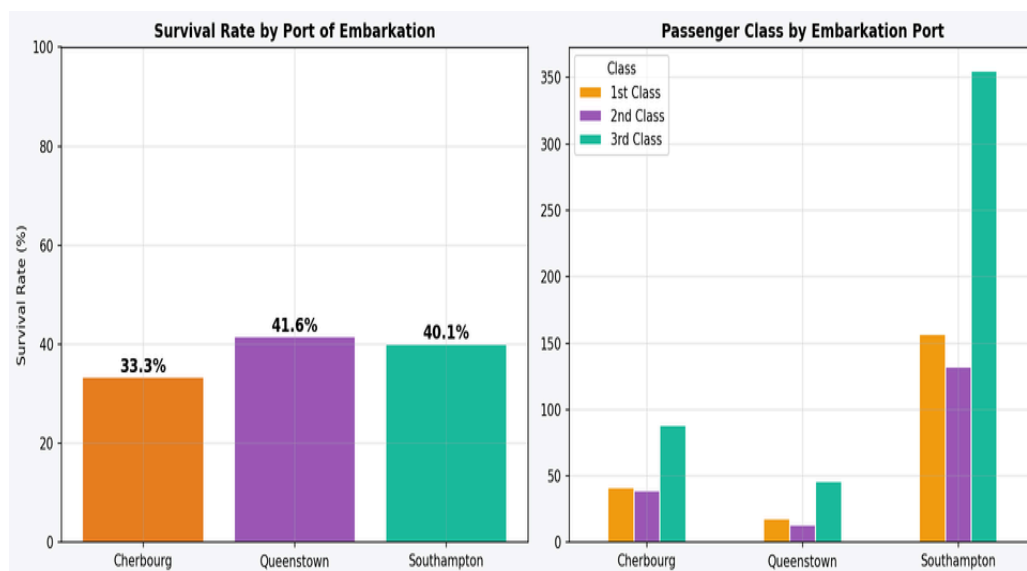


Figure 8: Survival rate by embarkation port (left) and class composition by port (right)

■ **Observation:** Cherbourg passengers had the highest survival (~55%) because a higher proportion were 1st-class. Southampton boarded the most passengers (644) and the most 3rd-class passengers, resulting in the lowest survival rate (~34%). The embarkation port effect is largely a proxy for passenger class — its importance in modelling may reduce once Pclass is included.

10. Multi-Feature Relationships



Figure 9: Multi-feature scatter plots — Age, Fare, and Pclass coloured by survival

■ **Observation:** The Age vs Fare scatterplot reveals a cluster of high-fare, middle-aged survivors — consistent with wealthy 1st-class passengers. Young adults paid across all fare ranges but show mixed survival driven by sex (not shown). The Fare distribution for survivors is distinctly right-shifted compared to non-survivors, reinforcing Fare's predictive value.

11. SibSp & Parch Analysis

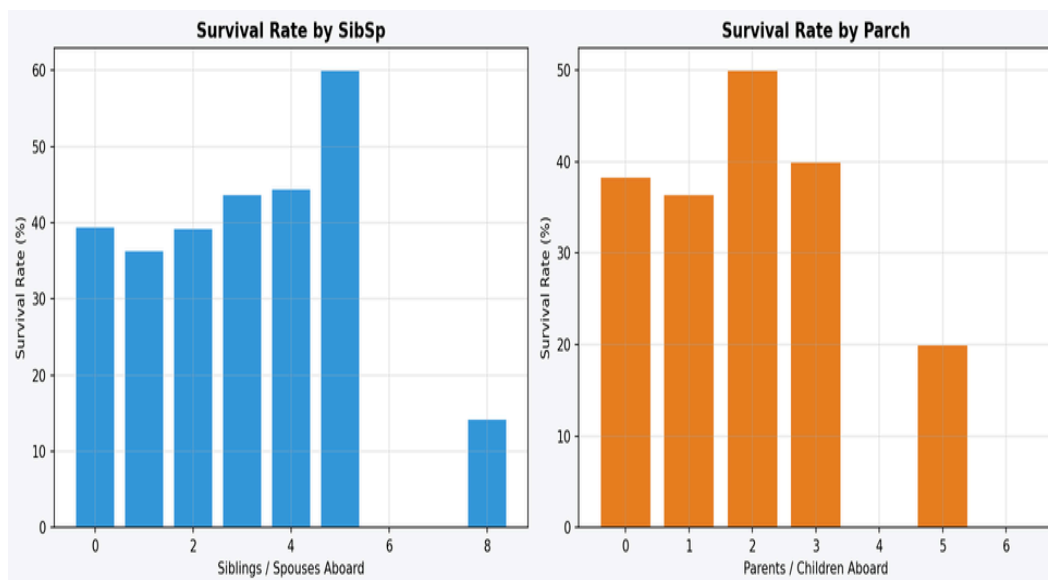


Figure 10: Survival rate by number of siblings/spouses (left) and parents/children (right)

■ **Observation:** Having 1 sibling or spouse aboard was associated with better survival — couples may have helped each other. Passengers with 5–8 siblings/spouses showed very low survival — these large sibling groups likely correspond to poor 3rd-class families. For Parch, 1–3 parents/children showed reasonable survival, consistent with family evacuation priority. Large values of both features are unreliable estimates due to very small sample sizes ($n < 10$).

12. Summary of Findings & Recommendations

Key Findings

Rank	Feature	Finding	Impact on Survival
1	Sex	Females: 71.3% survival vs Males: 21.1%	Very High ↑
2	Pclass	1st: 63.0% → 2nd: 42.4% → 3rd: 26.9%	High ↑
3	Fare	Higher fare strongly linked ($r=0.26$)	High ↑
4	Age	Children (<16) had ~52% survival	Moderate ↑
5	FamilySize	Size 2–4 best; solo & 7+ worst	Moderate
6	Embarked	Cherbourg best (proxy for class)	Low (confounded)
7	Cabin	83% missing; HasCabin flag useful	Low

Feature Engineering Recommendations

- **Sex_encoded:** Binary encode: female=1, male=0 (strongest predictor)
- **FamilySizeGroup:** Bin into: Alone(1) / Small(2-4) / Large(5+)
- **HasCabin:** Binary flag: 1 if Cabin is known, 0 if missing
- **Age imputation:** Fill missing Age with median grouped by Sex + Pclass
- **Fare_log:** Apply $\log_{1p}(\text{Fare})$ to reduce extreme skew
- **Embarked fill:** Fill 2 missing values with mode 'S' (Southampton)
- **Title extraction:** Extract title from Name: Mr, Mrs, Miss, Master — strong proxy for age+sex

Modelling Recommendations

- Use F1-score and AUC-ROC (not accuracy) due to class imbalance (38.4% vs 61.6%).
- Start with Logistic Regression as a transparent baseline to quantify feature importance.
- Upgrade to Random Forest or Gradient Boosting (XGBoost/LightGBM) for best performance.
- Apply 5-fold stratified cross-validation to ensure robust evaluation.
- Feature importance from tree models will confirm the EDA ranking: Sex > Pclass > Fare > Age.
- Consider removing Cabin (after creating HasCabin), PassengerId, and Ticket from modelling.

METRIC	VALUE	METRIC	VALUE
Total Passengers	891	Overall Survival	38.4%
Female Survival	71.3%	Male Survival	21.1%
1st Class Survival	63.0%	3rd Class Survival	26.9%
Median Age	30.5 years	Median Fare	£14.2
Missing Age	19.9%	Missing Cabin	82.7%